

Modeling the complexity of tree growth with hierarchical Gaussian processes and structured shocks

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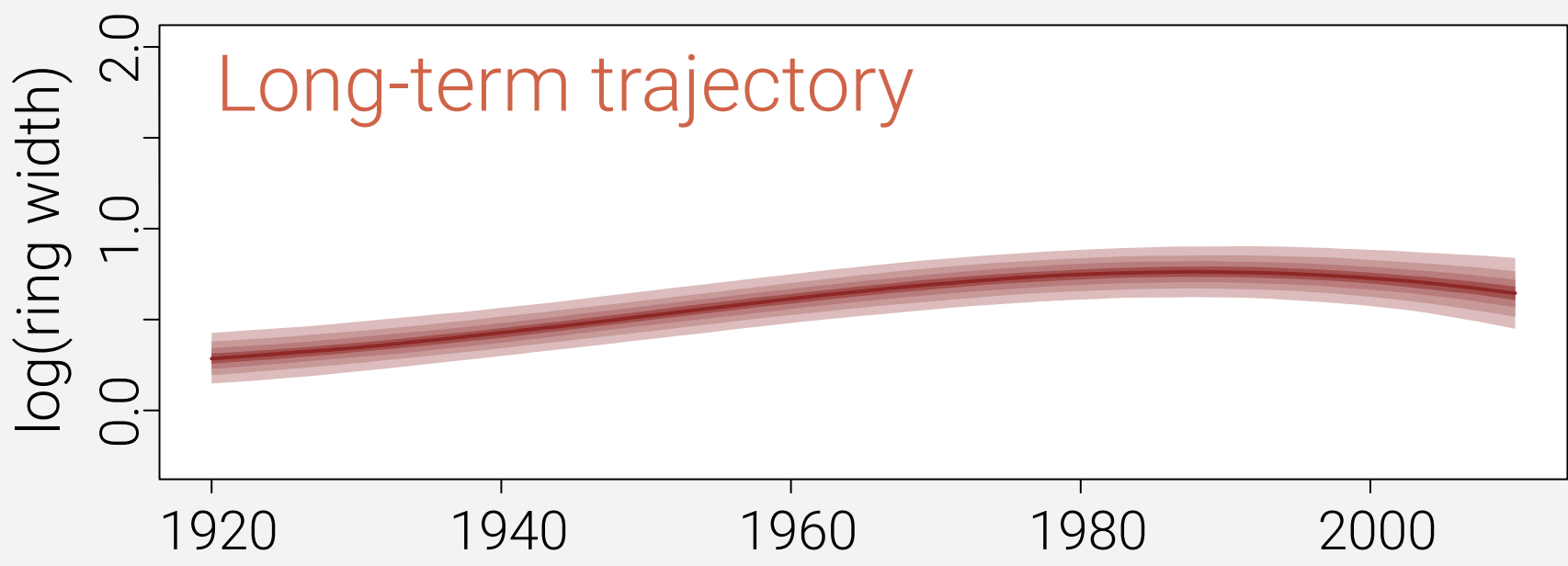
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OVERVIEW

- Climate change is **intensifying droughts** through shifts in temperature, vapor pressure deficit (VPD), and soil moisture [1]
- Combined effects of these drivers on tree growth remain poorly understood at large scales [2,3]
- A new **Bayesian approach** to jointly model tree growth across Western US, at the tree level
- Our model is able to separate climate effects from other complex dynamics (e.g. allometry)
- Some extreme drought caused **exceptional growth reductions** far exceeding what standard climate variables predict
- Current models likely **underestimate growth declines** during extreme events

1. MODELING GROWTH AT THE TREE LEVEL

- Tree-ring width time series are **one of the longest and most spatially extensive** records of forest response to climate
- However, tree rings are affected by **many drivers other than climate**: age-size effects, competition, biotic disturbance...
- We model jointly **3 components**:



1. A **long-term GP** for processes such as allometric constraints:

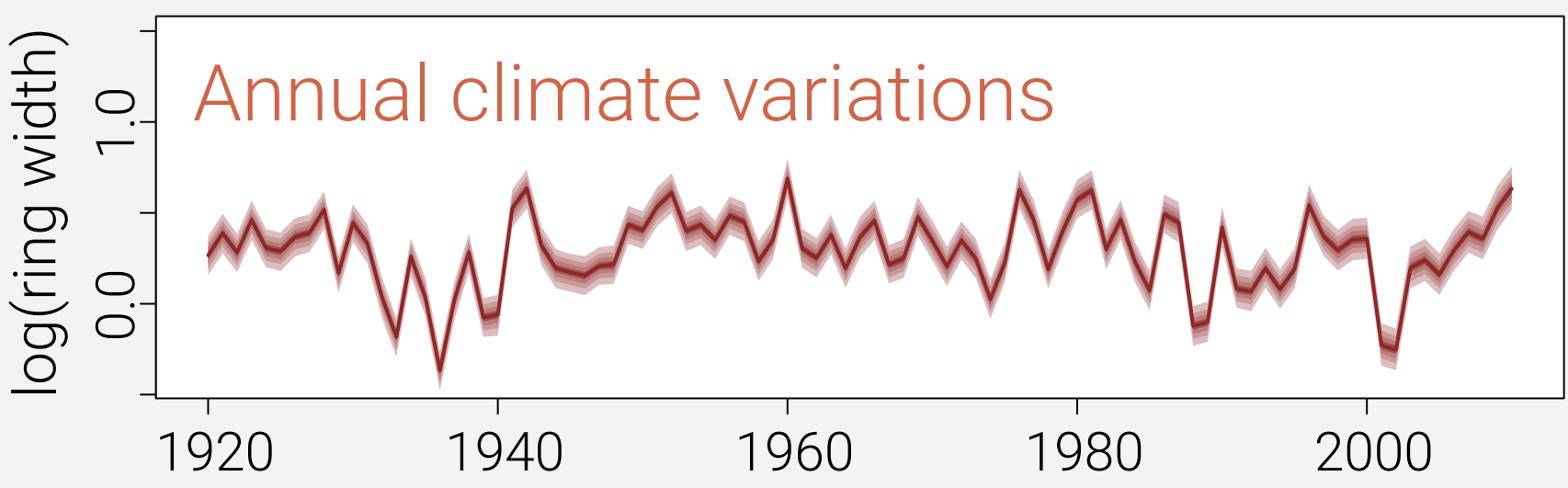
$$\vec{f}_{\text{long}} \sim \text{multinormal}(\vec{0}, \Sigma_{\text{long}})$$

characterized by an Gaussian kernel with species-specific length scale ρ_{species} and marginal variation γ_{species}

2. A **short- to mid-term GP** for processes such as neighbour competition dynamics:

$$\vec{f}_{\text{short}} \sim \text{multinormal}(\vec{0}, \Sigma_{\text{short}})$$

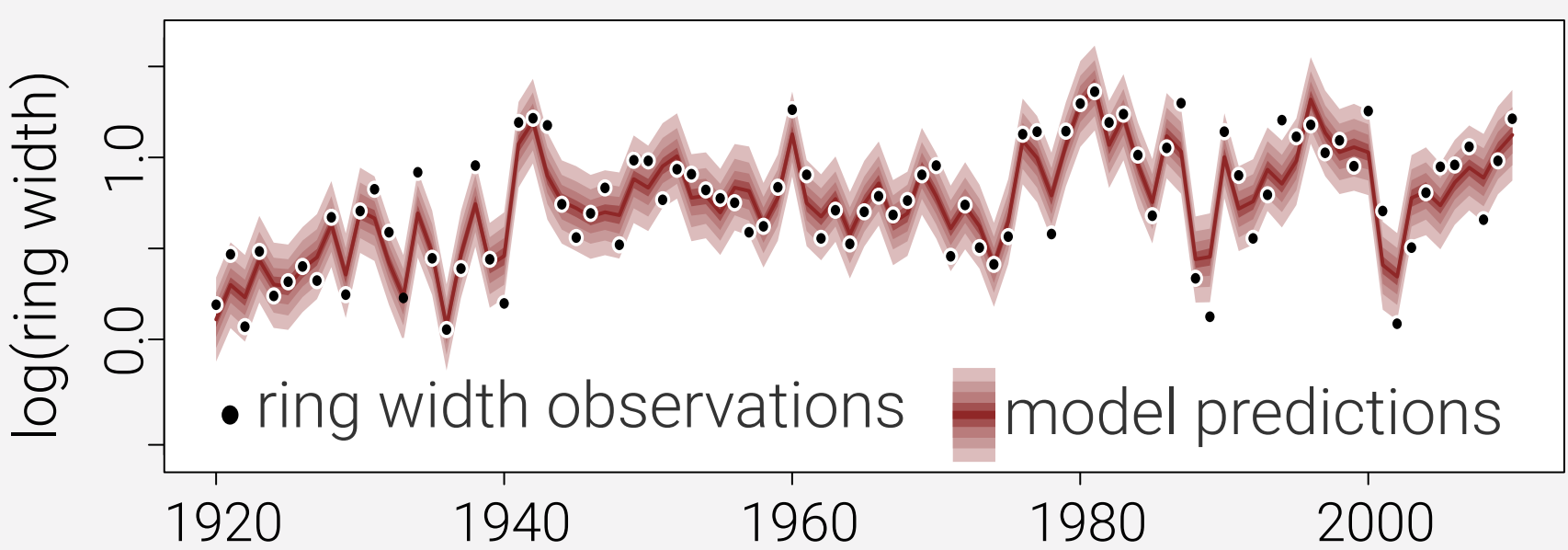
characterized by a second exponentiated quadratic kernel with length scale ρ_{short} and marginal variation γ_{short}



3. A **linear regression** of 3 climate variables with species-specific slopes $\vec{\beta}_{\text{species}}$

All together, we have, for each observation of ring width of each tree:

$$\log(y_i) \sim \text{normal}(\alpha + f_{\text{long},i} + f_{\text{short},i} + \vec{\beta}_{\text{species}} \cdot \vec{X}_{\text{clim},i}, \sigma)$$



2. EXTREME GROWTH REDUCTIONS (SHOCKS)

- But a tree can show **severe growth suppression**, far beyond what its climate response $\vec{\beta}_{\text{species}} \cdot \vec{X}_{\text{clim},i}$ predicts
- When one tree experiences a shock, **neighboring trees within the same forest** tend to as well, consistent with a **common external stressor** (rather than individual chance)

- We model this with a **nested mixture model** and latent states:

1. At the **forest level**, a stress can happen the year t with a probability that depends on the climate:

$$p(\text{forest-level event at } t) = \text{logit}^{-1}(\alpha_{\text{forest}}^{\text{stress}} + \vec{\beta}_{\text{stress}} \cdot \vec{X}_{\text{clim},t})$$

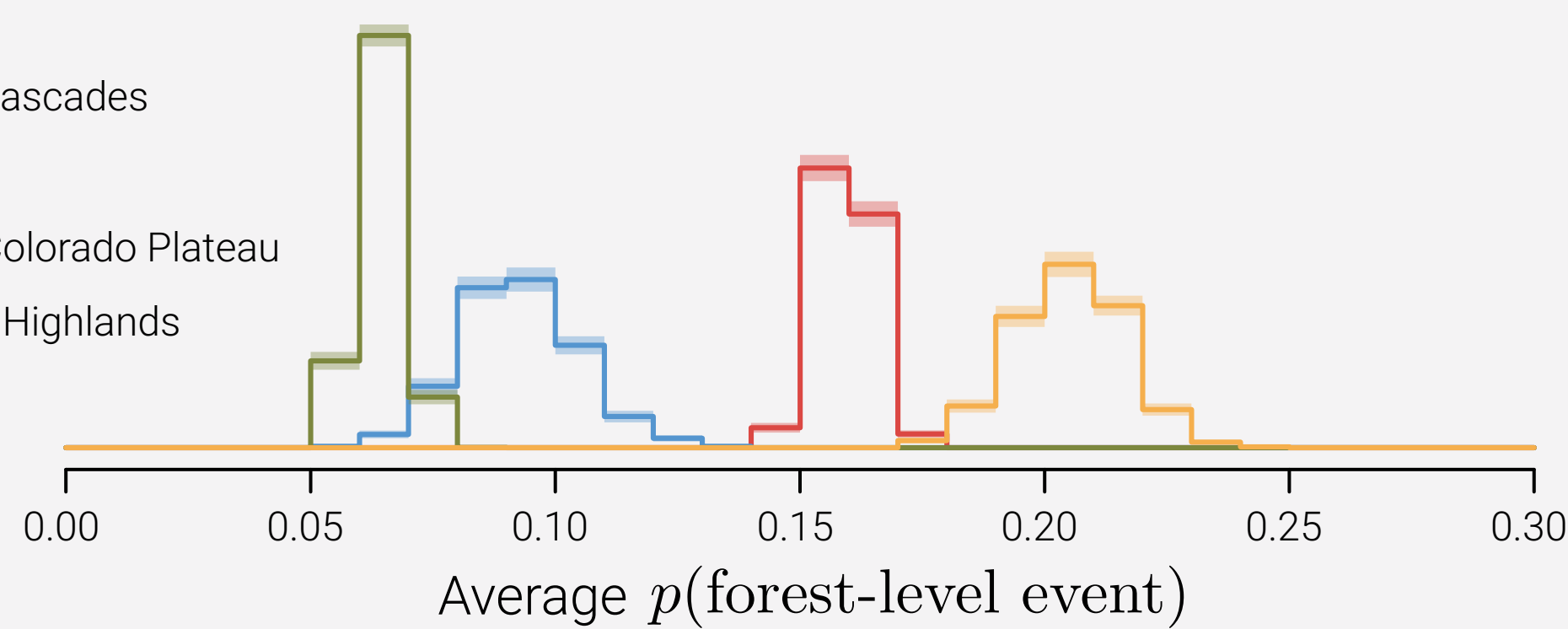
2. Given that a stress happens, a **tree** might experience a shock with some probability that depends on the species:

$$p(\text{tree shock} | \text{forest-level event at } t) = \text{logit}^{-1}(\alpha_{\text{forest}}^{\text{shock}} + \alpha_{\text{species}}^{\text{shock}})$$

If a shock happens, this can lead to a growth depression of a species-specific amplitude τ_{species} ... or even an entire shutdown

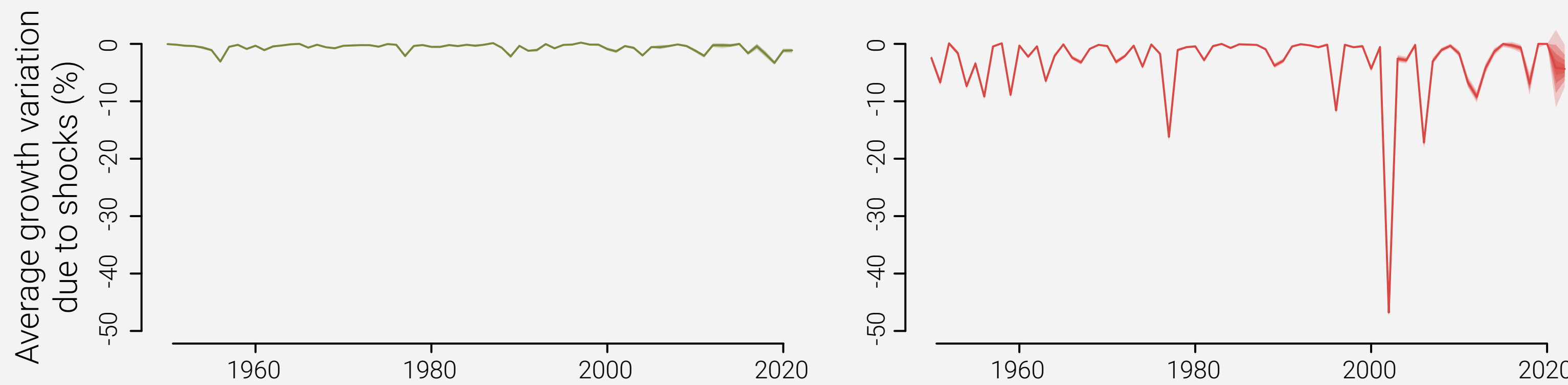
3. CONTRIBUTION OF SHOCKS TO REDUCED GROWTH

■ Northern Rockies & Cascades
■ Sierra Nevada
■ Southern Rockies & Colorado Plateau
■ Southwest Deserts & Highlands

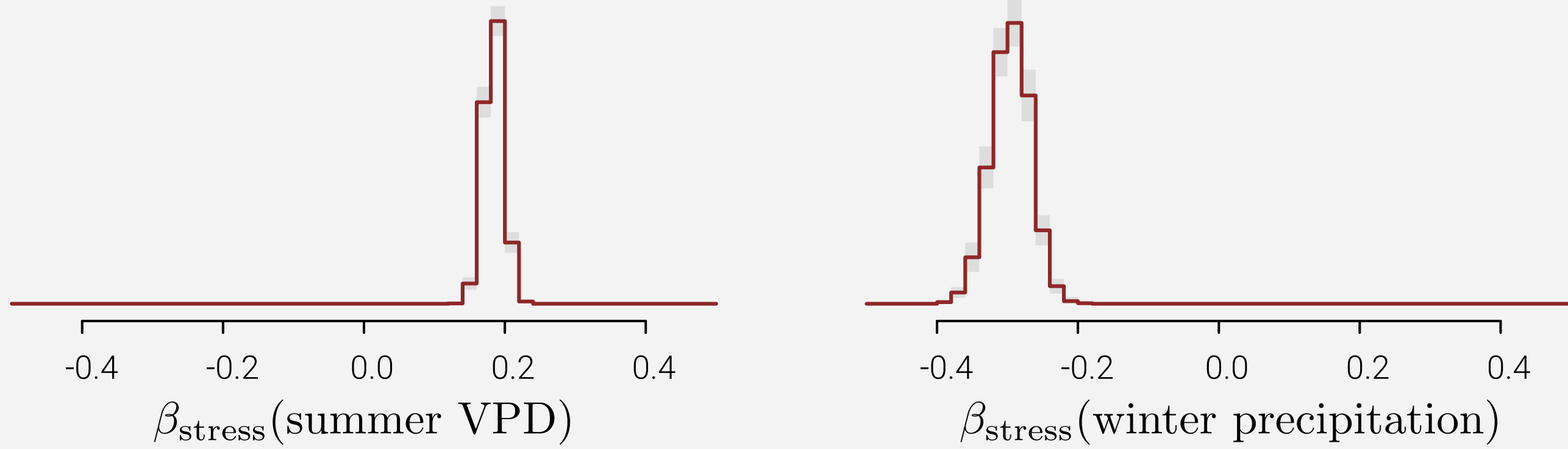


- Shocks are **more frequent** and have a stronger impact in **drier regions**

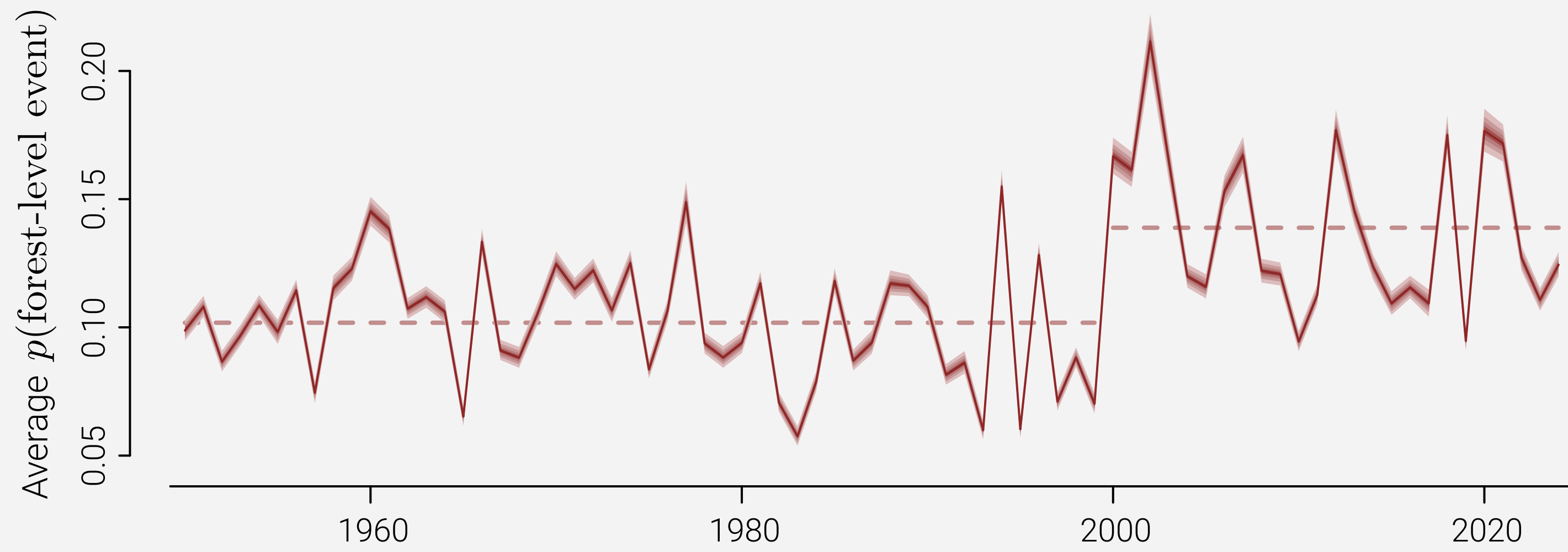
- The shocks during the 2002 drought reduced growth by **up to 50%**



4. CLIMATE AS A DRIVER OF GROWTH SHOCKS



- **Warm and dry summers** (high vapor pressure deficit, VPD) increase the probability of forest-level stress events, as well as **reduced winter precipitation** (limiting soil water recharge)



- Probability has **increased by 30%** across the western US after 2000 (10% to 14%)

5. IMPLICATIONS FOR FORESTS AND FORECASTS

- Classical dendroclimatology extracts climate signals by averaging out individual variation (top-down, [4]), while here **we model individual tree growth** (bottom-up)
- Accounting for these **non-linear, individual-level responses is critical for accurate forecasts** of forest vulnerability and carbon storage under climate change [5]

References

- [1] Gebrechorkos et al. Nature (2025)
- [2] Babst et al. Sci Adv (2019)
- [3] Gao et al. Nat Ecol Evol (2022)
- [4] Melvin & Briffa. Dendrochronologia (2008)
- [5] Anderegg et al. Nat Clim Change (2020)